

Macroeconomic Forecasting: Forecasting E-Commerce Retail Sales

Shivani Samlal

I. Project Overview

This project analyzes the time series behavior and constructs a forecasting model of U.S. e-commerce retail sales, measured by the *Advance Monthly Retail Trade: Retail E-Commerce Sales (ECOMNSA)* series, which was obtained from the Federal Reserve Economic (FRED) database. The series comprises quarterly data spanning from 1999 to 2025. The model will aim to produce a four-quarter, or one-year, forecast of future e-commerce sales.

II. Exploratory Data Analysis

The data was plotted in levels to better understand the underlying trends. The time series plot illustrates a consistent upward trend over time, revealing non-stationarity in levels (*Figure 1*). The series does not fluctuate around a constant mean and demonstrates periods of volatility. There is a particular heightened spike after the first quarter in 2020. This is reflective of possible macroeconomic structural changes or external shocks within the sample period. To formally test for stationarity, an Augmented Dickey-Fuller (ADF) test was conducted with an intercept and a deterministic trend. The results failed to reject the null hypothesis of a unit root (-1.2675 at $p = 0.89$), which indicated non-stationarity and that transformations are needed.

The series was transformed using raw first differencing in an attempt to stabilize the variance and remove the trend, which was followed by a log transformation. The ADF test was performed on both series to confirm stationarity. The log-differenced data allows the series to exhibit a smaller, stable variance centered around zero (*Figure 1*).

Seasonality was checked to further examine inherent patterns within the dataset, as it was not visible due to the persistent upward trend. A boxplot of e-commerce sales was constructed and revealed

stark value differences throughout the quarters (*Figure 2*). The dominance of quarter four sales is consistent with amplified end-of-the-year sales in the market. To further confirm, the autocorrelation function displays significant spikes at lags 4, 8, 12, and so on – in multiples of four, suggesting seasonality. Notably, seasonality persists even after log-differencing.

The autocorrelation function (ACF) of the level series shows a slow decay across all lags, which reflects non-stationarity. On the other hand, the log-differenced series illustrates an oscillating pattern, but the autocorrelation remains in seasonal lags. The partial autocorrelation function (PACF) shows significance in earlier lags, particularly lag 1 and lag 3. The lack of a pronounced cutoff in both the ACF and PACF implies that the series is a mixed structure of both a moving average and autoregressive process.

An essential requirement for forecasting is covariance stationarity, which is defined by a constant mean and a covariance structure that will not change over time. The level series violates these two conditions, as it has an upward trend and increasing variance over the sample period. After log-differencing, the series has a constant mean and variance, indicative that the trend component was successfully removed. However, there is still significant autocorrelation within seasonal lags, particularly at lag 4. Therefore, it is not fully covariance stationary due to the dynamics of the underlying seasonal structure. In addition, seasonality makes up a vast part of the explanation of the data; therefore, it would be ignorant to just ignore it. For e-commerce sales, the seasonality that is present is stochastic, as it is persistent over time and has dependence on past values. As a result, seasonal effects can be modeled through lag structures or a full seasonal ARIMA specification rather than fixed seasonal dummy variables.

III. Model Strategy and Candidate Models

The objective of this analysis is to construct a parsimonious forecasting model for e-commerce sales for the following year. After the exploratory analysis, the series is transformed using a logarithmic transformation and first differencing, which resulted in a growth rate series that exhibits more stable

properties. The transformation of $dlny_t$ to $\Delta \ln(y_t)$ allowed the series to be modeled as covariance stationary.

Following the observation of the autocorrelation structures of the transformed series suggests a mixed autoregressive and moving-average structure. This consideration is due to the fact that the ACF does not cut off sharply and the PACF exhibits gradual decay – importantly, there is significant correlation at lag 4 because of the quarterly seasonality. These models are designed to capture both the dynamics in the short run and the seasonal structures. The baseline model specification would look like:

$$dlny_t = \beta_0 + \phi_1 dlny_{t-1} + \phi_4 dlny_{t-4} + \epsilon_t$$

Within this model specification, the first lag captures the short-run dynamics in growth rates, while the fourth lag captures the dependence of seasonality correlating to the quarterly frequency. The inclusion of the seasonal lag successfully reflects the autocorrelation structure and dependence of the same quarter in the previous year.

The h-step ahead model can be written as

$$dlny_{T+h} = \beta_0 + \phi_1 dlny_{T+h-1} + \phi_4 dlny_{T+h-4} + \epsilon_{T+h}$$

The associated point forecast would be written as:

$$dln\hat{y}_{T+h|T} = \hat{\beta}_0 + \hat{\phi}_1 dlny_{T+h-1|T} + \hat{\phi}_4 dlny_{T+h-4|T}$$

However, to properly evaluate model robustness, additional autoregressive specifications are to be considered. In particular, higher-order autoregressive models are to be estimated to capture other complexities. Therefore, a grid search over a small set of autoregressive specifications is conducted. Models are estimated with combinations of short-run lags $\in \{0, 1, 2, 3\}$ and an optional seasonal lag at four.

$$dlny_t = \beta_0 + \sum \phi_i dlny_{t-i} + \epsilon_t$$

While higher-order AR models improved the fit of the series, diagnostic testing captured serial autocorrelation in the residuals, meaning the specification was incomplete. To address these limitations, the model was extended to include a moving average component. Analysis was conducted on combinations of lags $\in \{0, 1, 2, 3\}$. Resulting in the following ARIMA models:

$$dlny_t = \beta_0 + \sum \phi_i dlny_{t-i} + \sum \theta_i \epsilon_{t-i} + \epsilon_t$$

These specifications are evaluated based on the Akaike Information Criterion (AIC), the Schwartz Information Criterion (SIC/BIC), statistical significance of the coefficients, and diagnostics of the residuals. More importantly, out-of-sample forecast accuracy will be conducted between the top two parsimonious models.

IV. Model Selections and Diagnostic Evaluation

To choose the most parsimonious and fitted model specification for forecasting, two selected models were evaluated: a higher-order autoregressive model with a seasonal lag and an ARIMA model that has both autoregressive and moving average components embedded with seasonal persistence.

The first specification is a higher-order autoregressive model that includes short-run dynamics and quarterly seasonal lags:

$$\text{Model 1: } dlny_t = \beta_0 + \phi_1 dlny_{t-1} + \phi_2 dlny_{t-2} + \phi_3 dlny_{t-3} + \phi_4 dlny_{t-4} + \epsilon_t$$

This model was selected as the best autoregressive structure based on a grid search over lag lengths. This specification exhibits strong in-sample fit, as the AIC was estimated to be -304 and the BIC to be -301. The coefficients are negative and statistically significant in the first three lags; on the other hand, the last lag is positive and statistically significant.

Albeit the strong in-sample performance, diagnostic testing exhibits misspecification. The residual correlogram shows significant autocorrelation, and the Q-statistics reject the null hypothesis of no serial correlation. In addition, the residuals do not behave like white noise. The results reveal that the autoregressive model fails to capture the entirety of the underlying dynamics of e-commerce sales. The persistence of residual autocorrelation suggests omitted dynamics, which is likely in the form of moving average components.

To address these concerns, the model is extended to include a moving average term, seen in the following ARIMA model specification:

$$\text{Model 2: } dlny_t = \beta_0 + \phi_1 dlny_{t-1} + \phi_2 dlny_{t-2} + \phi_3 dlny_{t-3} + \phi_4 dlny_{t-4} + \theta_1 \epsilon_{t-1} + \epsilon_t$$

This specification incorporates short-run dynamics through lags 1-3, persistence of seasonality through lag 4, and the contribution of shocks through the MA(1) structure.

The ARIMA model produces statistically significant coefficients across all coefficients. The first three autoregressive terms remain negative and significant. This indicates short mean reversion on growth rates. The fourth lag is still positive and significant, which solidifies the presence of quarterly seasonality. The moving average term is positive and statistically significant, which is reflective of past shocks contributing to current values. The model's information criteria are similar to the previous model – the AIC is approximately -308 and the BIC is -287.

The residual diagnostics proved to fare well, as the residual autocorrelation has no significant spikes. The Q-statistics fail to reject the null hypothesis of no serial correlation, and the residuals behave like white noise. Therefore, the ARIMA(4,1,1) specification is selected as the preferred model.

V. Forecasting Performance

Using the ARIMA(4,1,1) model specification, a one-year-ahead forecast, or four quarters, of e-commerce retail sales were predicted. The forecast is immediately generated after the last quarter. As seen in *Figure 4*, the baseline ARIMA forecast suggests that e-commerce sales will continue to grow over the next year. However, the sales are at a more considerable pace compared to heightened increases in previous quarters. This is intuitive, as the model relies on past underlying dynamics, which smooth volatility in later periods and produce careful predictions.

To accommodate for uncertainty, two alternative scenarios were constructed (*Figure 5*). The “bull case” assumes optimistic higher-than-expected growth, reflecting the continued expansion and sustained demand in online e-commerce. This scenario expects e-commerce sales to increase rapidly compared to our baseline prediction. On the other hand, the “bear case” assumes a slower growth in sales, probably due to normalization in online consumption post-COVID-19 surge in 2020. Although it produces a lower trajectory, it still remains an upward trend.

All three predictions are an upward trend, reflecting that there is a long-run growth pattern in e-commerce sales. The difference between the “bear and bull cases” continues to highlight the uncertainties in future environments. This range aids in providing multiple scenarios and enhances the robustness of the possible positive or negative risks to potential e-commerce sales. However, the single-point forecast illustrates a stable, upward trend acknowledging the underlying trend in the data.

VI. Policy Implications

The continued upward trend in e-commerce sales emphasizes the importance of digitalization. Policymakers should prioritize investments in cybersecurity and digital pay platforms to aid in supporting the sustained growth in e-commerce activity.

In addition, the growth in e-commerce retail sales has heavy implications for labor markets. There may be a possible shift away from traditionalistic labor markets toward digital logistics and technology-driven employment. Officials would need to support these workforce transitions by creating education initiatives or accessible training programs.

Overall, the forecasting analysis illustrates that the long-term trajectory of e-commerce sales is positive, but there is still uncertainty around short-run fluctuations, which may warrant cautious policy responses.

Appendix

Figure 1: E-Commerce Retail Sales and Data Transformations

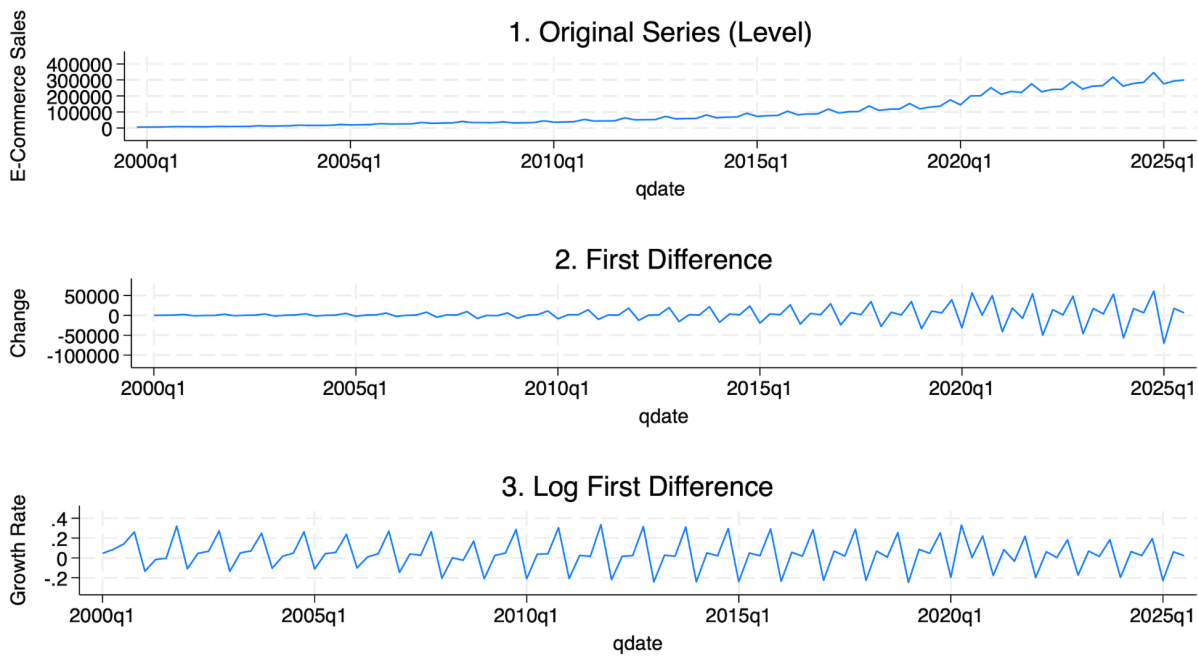


Figure 2: Seasonality Boxplot Check: E-Commerce Sales by Quarter

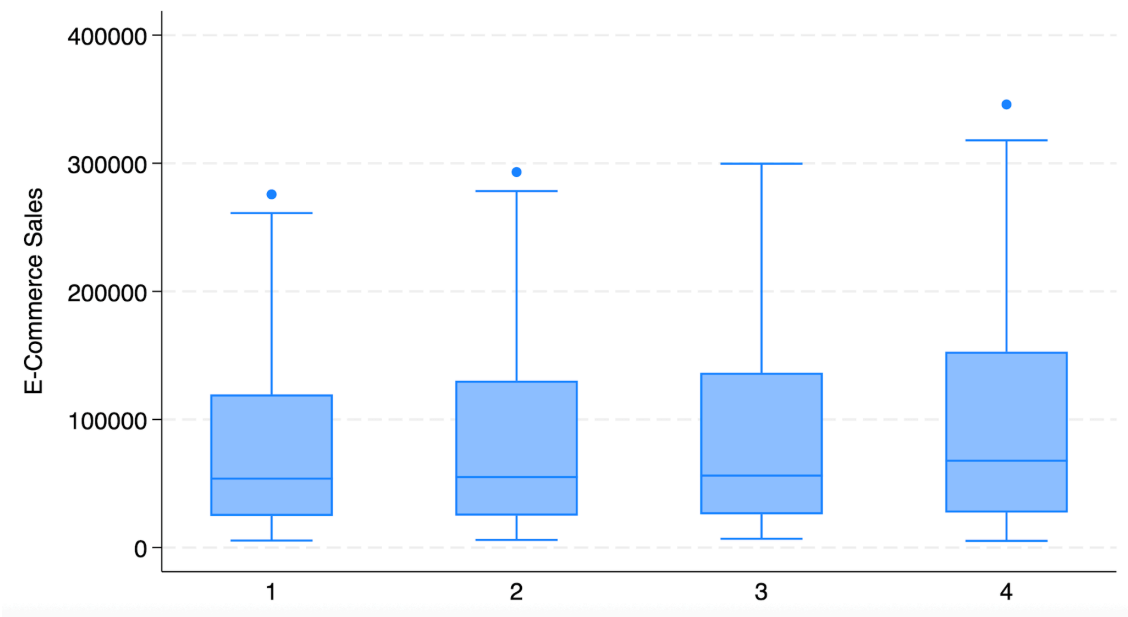


Figure 3: ACF and PACF plots

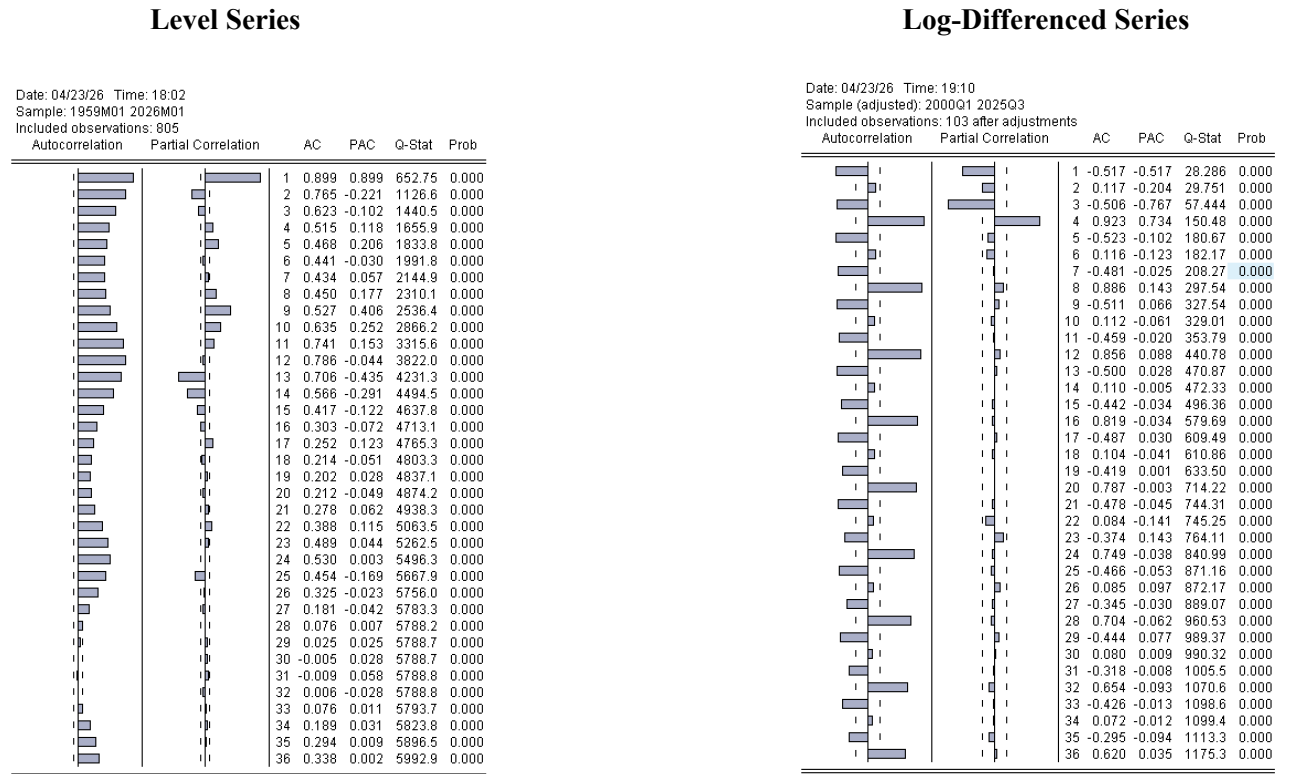


Figure 4: Final E-Commerce Forecast

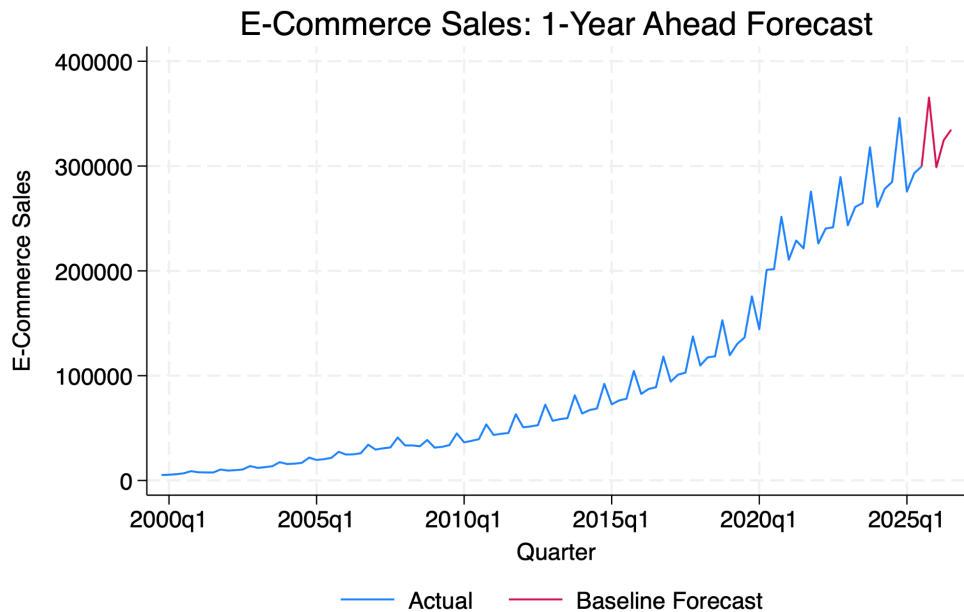


Figure 5: Forecast Scenario Analysis

